

Berkay Guler

Irvine, California

Email: gulerb@uci.edu

[LinkedIn](#)

[GitHub](#)

Phone: +1 949 992 4830

[Website](#)

Professional Summary

I am a machine learning researcher specializing in wireless communications with a focus on developing novel methods for next-generation wireless systems. My research and professional experience spans machine learning applications in vision, language, and signal processing, grounded in strong fundamentals in electrical engineering and computer science.

Education

- University of California, Irvine** Sept. 2023 – June 2027
Ph.D. in Networked Systems, Computer Science, (CGPA: 3.82/4.0) Irvine, California
• Research on wireless channel understanding, advised by Prof. Hamid Jafarkhani
- University of California, Irvine** Sept. 2023 – June 2025
M.S. in Networked Systems, Computer Science, (CGPA: 3.82/4.0) Irvine, California
• Research on wireless channel understanding, advised by Prof. Hamid Jafarkhani
- École Polytechnique Fédérale de Lausanne (EPFL)** Feb. – Aug. 2022
Exchange Student in School of Computer and Communication Sciences, (GPA: 5.24/6.0) Lausanne, Switzerland
• Research on the assessment of deep learning-based deepfake detectors, advised by Prof. Touradj Ebrahimi
- Bilkent University** Sept. 2019 – June 2023
B.S. in Electrical Engineering, Summa Cum Laude (GPA: 3.82/4.0) Ankara, Turkey
• Specialized in signal processing & machine learning
• Research on AI for medical imaging

Experience

- UC Irvine, Center of Pervasive Communications and Computing (CPCC)** Irvine CA, USA
Graduate Student Researcher (*Part-time*) Sep. 2023 – Present
• Innovating in **wireless channel understanding** using **self-supervised** approaches to achieve **task-agnostic** performance that matches supervised task-specific baselines.
• Advancing the state-of-the-art in **deep learning-based channel estimation** for **SISO OFDM** systems using a novel adaptive **transformer-based** model.
• Developing data-driven wireless solutions for **channel characterization, beam management, CSI feedback, channel estimation, and channel prediction.**
- DataBoss Security & Analytics** Ankara, Turkey
Machine Learning Engineer (*Full-time*) Feb. – Sep. 2023
• Conducted R&D on **text summarization** and **text normalization** using encoder-only and encoder-decoder transformer-based models in **PyTorch**.
• Developed containerized **APIs** to host text summarization and text normalization inference endpoints in **Python** using **Flask** and **Docker**.
• Built a robust **document-AI** pipeline combining pretrained CNNs for layout detection with a custom **MLP classifier** trained on hand-crafted features (HOG, Gabor filters, edge descriptors) to parse multi-article newspaper PDFs into structured JSON, outperforming baseline solutions by a great margin.

Huawei R&D Center

Machine Learning Researcher/Engineer Intern (*Part-time*)

Istanbul, Turkey

Aug. 2022 – Feb. 2023

- Research on **active learning** with **Transformers** to improve sample efficiency of labeled address data for a named entity recognition module.
- Developed a data-mining tool in **Python** to expand a dictionary used for dictionary-based methods for address parsing.

Multimedia Signal Processing Lab, EPFL

Undergraduate Student Researcher (*Part-time*)

Lausanne, Switzerland

Feb. 2022 – Aug. 2022

- Developed an evaluation framework for deep learning-based deepfake detectors, evaluating the detector robustness under realistic perturbations.
- Performed robustness analysis of top-performing **deep learning** models from the DeepFake Detection Challenge, implementing and evaluating **EfficientNet-B7**, **3D-CNN**, and **ensemble-based architectures** for **face swap detection**.

ICON Lab, Bilkent University

Undergraduate Student Researcher

Ankara, Turkey

Mar. 2022 – Sep. 2022

- Studied effective ways for synthetic image generation with **Diffusion Probabilistic Models** (DPMs) for skin cancer and retina datasets for classification tasks.

Baykar Technology

Embedded AI Intern

Istanbul, Turkey

June 2021 – Sep. 2021

- Developed embedded **C/C++** applications on a **STM32F7-series microcontroller**, focusing on troubleshooting latency issues on the **CAN** bus for a gimbal module developed for UAV platforms.

Publications

- **B. Guler**, G. Geraci, H. Jafarkhani, "PilotWiMAE: Pilot-Native Representation Learning for Wireless Channels," **submitted to *IEEE Journal of Selected Topics in Signal Processing (JSTSP): Wireless Foundation Models for AI-native 6G and Beyond (2026)***, preprint: <https://arxiv.org/abs/2605.22856>
- **B. Guler**, G. Geraci, H. Jafarkhani, "PilotWiMAE: Wireless Channel Pilots Are All You Need," **submitted to *ICML 2026: Workshop on AI and ML for Next-Generation Wireless Communications and Networking***
- **B. Guler**, G. Geraci, H. Jafarkhani, "A Multi-Task Foundation Model for Wireless Channel Representation Using Contrastive and Masked Autoencoder Learning," *IEEE Journal on Selected Areas in Communications (JSAC): Large AI Models for Future Wireless Communication Systems (2025)*, online: <https://ieeexplore.ieee.org/document/11455140>
- **B. Guler**, G. Geraci, H. Jafarkhani, "Robust Channel Representation for Wireless: A Multi-Task Masked Contrastive Approach," *NeurIPS 2025: Workshop on AI and ML for Next-Generation Wireless Communications and Networking*, (**oral presentation and best paper runner up**), online: <https://openreview.net/forum?id=KXNDs9ZGb9>
- **B. Guler**, H. Jafarkhani, "AdaFortiTran: An Adaptive Transformer Model for Robust OFDM Channel Estimation," *2025 International Conference on Communications (ICC)*, Montreal, Canada, online: <https://ieeexplore.ieee.org/document/11160810>
- **B. Guler**, B. Aygun, A. Gerek and A. S. Gurel, "Deep Active Learning for Address Parsing Tasks with BERT," *2023 31st Signal Processing and Communications Applications Conference (SIU)*, Istanbul, Turkey, online: <https://ieeexplore.ieee.org/document/10223996>

Graduate Level Coursework

- **Machine Learning & AI:** Deep Generative Models, Machine Learning, Image Analysis and Pattern Recognition
- **Communications:** Digital Communications I & II, Wireless Communications, Error Correcting Codes, Digital Signal Processing, Signal Processing for Communications
- **Mathematics:** Optimization, Random Processes, Statistics, Control Theory
- **Computer Science:** Graph Algorithms, Design and Analysis of Algorithms, Data Structures
- **Networks:** Computer Networks, Communications Networks, Network and Distributed Systems Security

Skills

- **Programming Languages:** Python, MATLAB, C/C++, and Java
- **Frameworks & Tools:** Git, PyTorch, Hugging Face, Scikit-learn, Bash, Flask, Docker, Sionna, Deep-MIMO, QuaDRiGa
- **Machine Learning & AI:** Neural networks (Transformer, CNNs, RNNs, fully-connected networks), representation learning (latent-predictive models and input-predictive models), generative modeling (large language models, score-based (diffusion) models, VAEs, auto-regressive models)
- **Languages:** English (Fluent), Turkish (Native)

Professional Activities

Reviewer , Neural Information Processing Systems (NeurIPS) 2026, Sydney, Australia	2026
Program Committee , AI4NextG Workshop @ ICML 26', Seoul, South Korea	2026
Presenter , Information Theory and Applications Workshop, San Diego, USA	2026
Presenter , Neural Information Processing Systems (NeurIPS) 2025, San Diego, USA	2025
Presenter , IEEE International Conference on Communications (ICC 2025), Montreal, Canada	2025
Reviewer , International Conference on Learning Representations (ICLR 2026)	2025
Reviewer , IEEE Journal of Selected Areas in Communications (JSAC)	2025
Reviewer , Data Compression Conference (DCC 2026)	2025
Mentor , Undergraduate Research Opportunities Program (UROP), UC Irvine	2024 – 2025
Mentor , Graduate International Connection, UC Irvine	2023 – 2024
Ambassador , Peers Offering Wellness Educations Resources (POWER), UC Irvine	2024 – 2025
Volunteer , Orange County Alumni Association	2024 – Present
Head of Sponsorship , Bilkent MUN Club	2021 – 2022
President , Bilkent Judo Club	2020 – 2021
Professional Memberships: IEEE Student Member, IEEE Communications Society Member	2025 – Present

Teaching Experience

Teaching Assistant , Bilkent University	Jan. 2020 – Jan. 2022
Tutored students in Introduction to Data Analysis in Python, Programming for Engineers, and Electricity and Magnetism courses	

Awards

- Best paper runner up award at AI4NextG Workshop at NeurIPS'25
- Recipient of Henry Samuelli Endowed Fellowship
- Recipient of Networked Systems Fellowship
- Received an Excellence Scholarship including full tuition waiver, dormitory fee waiver, and monthly

stipend during the entire bachelor's program

2019–2023

- Ranked **102nd**, **471st**, and **445th** in nationwide language, social sciences, and math/sciences university exams taken by more than two million students, respectively.

2018